

Indraprastha Institute of Information Technology Delhi, New Delhi

EEE ESC 207 A Ecology, Evolution and Environment

Mid - Semester Examination (01/03/2026)

MM 20

Time : 1 hr.

1. Answer True or False (Each question carries 0.5 mark)

- i. Tigon is a distinct species.
- ii. Some chemoautotrophic bacteria can also be facultative bacteria.
- iii. Bacteriophage is an anaerobic bacteria.
- iv. Microalgae can grow as ten inch long ropes in the sea.
- v. Sprulina is a blue green algae.
- vi. Experimental work at Yellowstone Park in USA showed that the presence of predators is not desirable in deer park.
- vii. Epilimnion zone is the lowest zone in water bodies.
- viii. The species which could not change under the changing environment did not perish.
- ix. Darwin was influenced by the work of Alexander von Humboldt who hailed from UK.
- x. Inbreeding depression refers to the decrease or loss of fitness and strength which is mainly caused due to inbreeding.
- xi. Uracil is a sugar phosphate backbone structure in the double helical structure of DNA.
- xii. Gene frequency is defined as the proportion of a population that carries a specific genetic variant, also known as allele frequency.
- xiii. Genetic drift or hitch hiking is the change in frequency of an existing gene variant in the population due to random chance.
- xiv. Bacillus have spherical or ovoid shape and appearance.
- xv. Homologous structures imply convergent evolution.
- xvi. rRNA (ribosomal RNA) brings amino acids and reads the genetic code during translation.

2. Fill in the blanks (Each question carries 0.5 mark).

- i. Lichens consist of and fungi.
- ii. Raw leafy green vegetables and produce that grow close to the ground, particularly cabbage, lettuce, spinach, and strawberries, are commonly contaminated with free-living like Acanthamoeba spp. and Vermamoeba vermiformis.
- iii. PFAS stand for substances/chemicals/pollutants.
- iv. Biomagnification is the increasing concentration of persistent, toxic substances—such as heavy metals (mercury) or pesticides (DDT)—in the tissues of at successively higher trophic levels in a food chain.
- v. Secondary wastewater treatment is mainly a Treatment process.
- vi. There is a five finger option or understanding of the five processes of evolution — small population, non-random mating, mutations, , adaptation .

- vii. Development of roses of different colour is an example of process
- viii. The most prominent small cow in India is the....., a dwarf zebu breed from Andhra Pradesh.
- ix. Many embryos are similar in appearance during the early stages of development, suggesting that similar are at work.
- x. Analogous structures imply evolution (Similar function but different origin).
- xi. Structures that are reduced in size or function serve as clues to the organisms' evolutionary history are known as Structures.
- xii. Organisms pass on their structures to the offsprings was proposed by
- xiii. The transfer RNA, a necessary component of translation (the biological synthesis of new proteins in accordance with the genetic code), could have been a replicator molecule itself in the
- xiv. LUCA stands for
- xv. Heritable variations that may underlay some of the mechanics in developmental plasticity (.....) and canalisation (constant forms) under different environmental challenges.
- xvi. Phenotypic variance is often caused by a combination of factors.

Give 5-10 lines (pen lines) short answers

- 3. What are xynobots? 1
- 4. What is the role of genetic engineering in evolution? 1
- 5.What is crispr in genetic engineering? 1
- 6. What is chemogeny ? 1